

The Binomial Theorem



Binomial coefficients

- 1 Evaluate each binomial coefficient:

a $\binom{6}{2}$

b $\binom{8}{0}$

c $\binom{5}{5}$

d $\binom{10}{3}$

- 2 Write down row 4 of Pascal's triangle (the row beginning 1, 4, ...) and explain how it relates to the expansion of $(a + b)^4$.

Expansions and specific terms

- 3 Expand $(x + 2)^4$ fully.
- 4 Find the coefficient of x^3 in the expansion of $(2x - 1)^5$.
- 5 Find the term containing y^4 in the expansion of $(x + y)^7$.
- 6 Use the binomial theorem to expand $(1 + 0.02)^5$ to the x^2 -style second-order term, and hence estimate 1.02^5 to four decimal places.

More expansions and terms

- 7 Expand $(2x - y)^3$ fully.
- 8 Find the middle term of the expansion of $(x + 3)^6$ (the term with x^3).
- 9 Find the constant term in the expansion of $(x^2 + \frac{1}{x})^6$.

Sums of binomial coefficients

- 10 Evaluate $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$ without computing each term, using the identity $\sum_{k=0}^n \binom{n}{k} = 2^n$.
- 11 Use the fact that $\binom{n}{k} = \binom{n}{n-k}$ (symmetry of Pascal's triangle) to explain why $\binom{9}{2} = \binom{9}{7}$ without computing either value directly.

Further expansions

- 12 Find the coefficient of x^4y^3 in the expansion of $(x + 2y)^7$.
- 13 Find the first three terms in the expansion of $(1 - 2x)^6$ in ascending powers of x .
- 14 Use the binomial theorem with $(1 - 0.01)^4$ to approximate 0.99^4 to four decimal places, using the first three terms.
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Deriving Pascal's triangle recursively

- 15 State Pascal's Rule, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, and use it to compute $\binom{6}{3}$ from row 5 of Pascal's triangle (1, 5, 10, 10, 5, 1).
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The general term formula

- 16 State the general term (the $(k + 1)$ th term) formula for the expansion of $(a + b)^n$, and use it to find the 5th term of $(x - 2)^8$ directly (without expanding the whole thing).
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Combinatorial interpretation of binomial coefficients

- 17 Explain the combinatorial meaning of $\binom{n}{k}$, and use it to explain why $\binom{n}{0} = \binom{n}{n} = 1$ for any n .
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Applying the binomial theorem to probability-style counting

- 18 A fair coin is flipped 5 times. Using $\binom{5}{k}$, find the number of outcomes with exactly 3 heads, and state the total number of possible outcomes (using 2^5).
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A closing challenge problem

- 19 Find the sum of all the coefficients in the expansion of $(3x - 2)^4$, using the fact that setting $x = 1$ gives the sum of all coefficients.
- 20 Find the coefficient of x^5 in the expansion of $(x^2 + 3)(x - 1)^6$, by finding the relevant terms from $(x - 1)^6$ first.
- 21 Use the binomial theorem to expand $(x + h)^3 - x^3$ fully, then divide by h and simplify — this expression is exactly the difference quotient used to find the derivative of $f(x) = x^3$ in calculus.

22 Evaluate $\binom{7}{2}$ and $\binom{7}{5}$, and confirm they are equal using the symmetry property.